

Determination of Plank's constant using LEDs

Aim:- To determine the value of Plank's constant by from the turn-on voltage of LEDs.

Apparatus: LED, Power supply and Voltmeter

Principle:

This experiment provides a simple method to determine the Planck's constant h , fundamental constant of quantum physics.

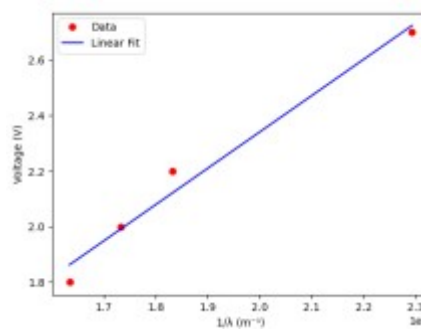
Light-emitting diodes (LEDs) emit photons when electrons recombine with holes across the p-n junction. When the forward voltage across the diode exceeds a threshold value V_0 (turn-on voltage) photons of energy $= eV_0$ are emitted from the LED. At this voltage, electrons gain enough energy to produce photons of a specific wavelength.

$$\text{Energy of emitted photon} = eV_0 = hf = \frac{hc}{\lambda}$$

eV_0 is the electrical energy supplied to an electron.

hf is the energy of the emitted photon of frequency

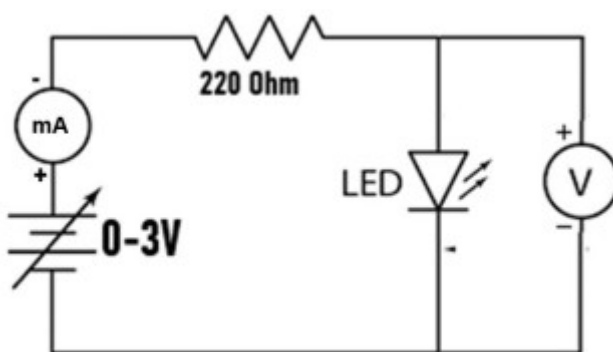
$$V_0 = \frac{1}{\lambda} \frac{hc}{e}$$



Plot a graph of $(1/\lambda)$ vs V_0 for different colours of LEDs, and perform a straight line fit to get the slope.

$$h = \frac{e}{c} \times \text{Slope}$$

Procedure:



Set up a circuit as shown in the diagram. Slowly increase the forward voltage (V) until the LED just begins to glow. Note this “turn-on” or threshold voltage, V_0 . To identify the exact turn-on point, observe the LED from the top of its casing and create a dark background using your hand or a piece of paper. This will help you detect the first visible light more accurately.

Observations:

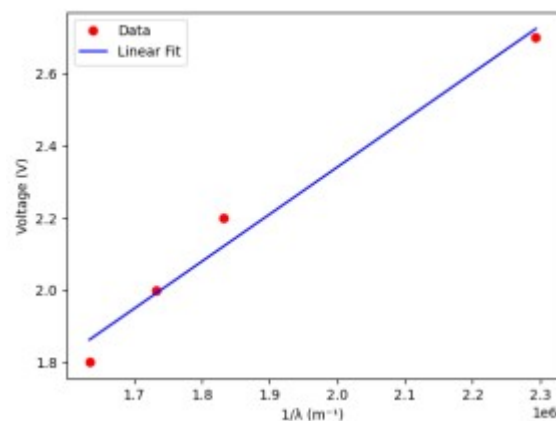
Note the specific wavelength for each LED (often provided by the manufacturer or measured with a spectrometer).

LED	λ ($10^{-9}m$)	V_0 (Volts)	$h = \frac{eV_0\lambda}{c} Js$
Red	612	1.8	5.87×10^{-34}
Yellow	577		
Green	546		
Blue	436		

$$h = \frac{eV_0\lambda}{c} = \frac{1.6 \times 10^{-19} \times 1.8 \times 612 \times 10^{-9}}{3 \times 10^8} = 5.87 \times 10^{-34} Js$$

Plot the threshold voltage V_0 on the y-axis against the inverse of the wavelength ($1/\lambda$) on the x-axis. Determine the slope of the line of best fit and Calculate Planck's Constant:

$$h = \frac{e}{c} \times Slope$$

**Results:**

The value of Planck's constant, $h = \dots\dots\dots Js$